The Voltaire Energy Terminal

A complete, plain-English guide to what this project is, why it exists, and exactly how it works

Written so that someone who has never heard the words "futures," "spread," or "backwardation" can finish this document and understand the entire project. The focus is the financial markets — what the numbers mean and why a trader cares — not the software.

How to read this document

This report is built in layers. We start with the absolute basics of the oil market — assuming you know nothing — and only once those are solid do we explain what the project actually does with them. If you already know what a "calendar spread" is, you can skim Part II; if you don't, Part II is where everything later will start to make sense.

- **Part I** — Why oil is traded at all, and who trades it.
- **Part II** — The building blocks: futures, the forward curve, spreads, volatility, inventories. *(The vocabulary. Read this slowly.)*
- **Part III** — What the project is, in one picture.
- **Part IV** — Phase 1: the live market dashboard.
- **Part V** — Phase 2: reading the market's "mood" and finding mispriced trades.
- **Part VI** — Phase 3: testing the trading idea on real data (the backtest).
- **Part VII** — How to read the results honestly.
- **Glossary** — Every term, defined in one line.

Part I — The oil market, from zero

What is being traded?

Crude oil is the raw material that gets refined into the fuels the world runs on: petrol (gasoline) for cars, diesel for trucks, jet fuel for planes, heating oil for homes. Because almost every economy depends on it, the price of oil is one of the most-watched numbers in finance.

But there is no single "price of oil." Oil comes in different grades from different places, and — crucially — it is bought and sold not just for *today*, but for delivery *months and years into the future*. Understanding the project requires understanding that second point deeply, so we will spend time on it.

The two benchmark oils: Brent and WTI

When people say "the oil price," they almost always mean one of two **benchmark** grades:

- **Brent** — oil from the North Sea (off the coast of the UK/Norway). It is the benchmark for most of the world (Europe, Africa, the Middle East). It trades on an exchange called **ICE**. In our data it carries the code `CO`.
- **WTI** (West Texas Intermediate) — oil produced in the United States. It is the benchmark for the Americas. It trades on an exchange called the **CME/NYMEX**. In our data it carries the code `CL`.

The two are similar but not identical, and the *gap* between their prices (the **Brent–WTI spread**) is itself one of the most-traded relationships in the market. We will return to this.

Who trades oil, and why?

Three broad groups:

1. **Hedgers** — airlines, refiners, oil producers. They trade to *reduce* risk. An airline that knows it must buy jet fuel next year can lock in a price today, so a price spike doesn't bankrupt it.
2. **Speculators** — hedge funds, trading desks, individuals. They trade to *make money* from price moves. They take on the risk the hedgers want to shed.
3. **Physical players** — companies that actually move barrels of oil around the world.

This project is written from the point of view of a **speculator / trading desk**: it is a tool to find and test profitable trades. It does not move real barrels.

Part II — The building blocks

This is the most important section. Every later part builds on it.

1. Futures contracts

You cannot easily buy a tanker of oil and store it in your garage. So the market trades **futures contracts** instead.

A **futures contract** is a standardized, legally binding agreement to buy or sell a fixed amount of oil at a fixed price, for delivery on a specific future date. One standard oil contract covers **1,000 barrels**.

The key idea: a futures contract has a **price today** even though delivery is in the future. That price moves up and down every second as people buy and sell, exactly like a stock. Most speculators never take delivery — they buy a contract, and sell it later at a (hopefully) better price, pocketing the difference. They are trading the *price*, not the physical oil.

> **Why this matters for the project:** everything the project analyzes is the *price of futures contracts*, captured as it moves through the day.

2. Contract months ("tenors")

Here is the part that surprises newcomers. For any one oil (say WTI), there isn't one futures contract — there is a whole **ladder** of them, one for each future delivery month, stretching out years.

Each contract is labelled by its delivery month using a letter code plus a year:

Code	Month	Code	Month
F	January	N	**July**
G	February	Q	**August**
H	March	U	**September**
J	April	V	October
K	May	X	November
M	June	Z	December

So `CL_N26` means "WTI, July 2026 delivery." `CO_Q26` means "Brent, August 2026 delivery."

The contracts are ranked by how soon they deliver:

- **M1 ("the front month" / "prompt")** — the nearest contract to expiry. The most actively traded.
- **M2** — the next one out.

- **M3** — the one after that. And so on.

In June 2026, for WTI, M1 = July (`N26`), M2 = August (`Q26`), M3 = September (`U26`).

3. The forward curve

If you line up the prices of all those contracts — M1, M2, M3, ... out into the future — and plot them, you get the **forward curve** (also called the *term structure*). It answers: "what does the market think oil will cost at each future date?"

The *shape* of this curve is enormously informative, and it comes in two flavours:

- **Backwardation** — near-term contracts are **more expensive** than far-term ones; the curve slopes **down**. This usually signals that oil is **scarce right now** — buyers are paying a premium to get barrels immediately. Backwardation is generally a "tight market" / bullish sign.
- **Contango** — near-term contracts are **cheaper** than far-term ones; the curve slopes **up**. This usually signals **oversupply** — there's plenty of oil now, so nobody pays extra for immediate delivery; in fact you'd rather buy later. Contango is generally a "loose market" / bearish sign.

> **Plain analogy:** imagine fresh strawberries. In peak season they're cheap now and you'd pay more for "guaranteed strawberries in winter" (contango). During a shortage, you'll pay a premium to get them *today* rather than wait (backwardation). Oil's curve tells the same story about scarcity.

4. Spreads — the heart of this project

A speculator can simply bet "oil will go up" by buying a contract. But that is a blunt, risky bet — it depends on the entire world oil price, which is driven by wars, economies, OPEC decisions, everything.

A more refined approach is to trade **spreads**: the *difference in price between two related contracts*. When you trade a spread you **buy one contract and simultaneously sell another**. You no longer care whether oil goes up or down overall — you only care about whether the *gap between the two* widens or narrows. This strips out the big, unpredictable market-wide moves and isolates a smaller, more stable relationship.

This project trades several kinds of spreads. Here are the ones it uses:

(a) Calendar spreads (M1–M2, M2–M3)

A **calendar spread** is the price difference between two contract months of the *same* oil. For example **WTI M1–M2** = (price of July WTI) − (price of August WTI).

- If you think near-term oil will get *tighter* (more backwardated), you **buy the spread** (buy the near month, sell the far month).
- The calendar spread is, in effect, a pure bet on the *shape of the forward curve* — on scarcity vs glut — without betting on the overall oil price at all.

(b) Butterflies ("flies")

A **butterfly** combines two calendar spreads to bet on the *curvature* of the forward curve — whether the middle of the curve is out of line with its neighbours. A WTI fly is built as:

> **+1 × M1 −2 × M2 +1 × M3**

(buy one July, sell two August, buy one September). If August is priced strangely relative to July and September, the fly captures it. It's an even more surgical, market-neutral trade than a calendar spread.

(c) The Brent–WTI spread

This is the price difference between the two benchmark oils: **Brent − WTI**. It reflects regional supply/demand differences (US vs the rest of the world), shipping costs, and pipeline bottlenecks. It is one of the most liquid and closely watched spreads in all of commodities.

(d) Crack spreads (background)

A **crack spread** is the difference between crude oil and the refined products made from it (petrol, diesel/heating oil). It approximates a refiner's profit margin — "buy crude, sell the fuel." The famous **3:2:1 crack** models a refinery turning 3 barrels of crude into 2 of petrol and 1 of diesel. *The project's broader framework analyzes cracks, but the live backtest data only contains crude, so the backtest itself trades crude spreads only — more on this later.*

5. Volatility

Volatility measures how violently a price swings around. High volatility = big, fast moves; low volatility = calm, gentle drift. It is usually expressed as an annualized percentage (e.g. "52% volatility"). Volatility matters because the *same* trade is far riskier in a stormy market than a calm one, and because some strategies only work in certain volatility conditions.

6. Inventories (stockpiles)

Governments and agencies report how much oil is sitting in storage tanks. In the US, the **EIA** (Energy Information Administration) publishes this weekly. Inventories are the single clearest read on supply and demand:

- **Stocks falling / below normal** → oil is being consumed faster than produced → **tight** market → supportive of prices.
- **Stocks rising / above normal** → a glut is building → **loose** market → bearish.

Analysts compare current stocks to the **5-year seasonal average** (because oil demand is seasonal — more driving in summer, more heating in winter), and express the gap as a number of **standard deviations** from normal (explained next).

7. The z-score and "mean reversion" — the two ideas the strategy rests on

These two concepts are the engine of the whole trading strategy, so they get extra care.

The z-score (how unusual is this number?)

Suppose a spread normally hovers around **1.50**, and on a typical day wiggles by about **0.10** up or down. One day it jumps to **1.80**. Is that a big deal?

A **z-score** answers exactly that. It measures *how far a value is from its own average, counted in "typical wiggles."* The "typical wiggle" is the **standard deviation**.

> z = (current value − average value) ÷ typical wiggle

In our example: (1.80 − 1.50) ÷ 0.10 = **+3.0**. The spread is 3 standard deviations above normal — a genuinely rare, stretched reading. A z-score near **0** means "totally normal." A z-score of **+2 or −2** means "unusually high / low." **±3** is extreme.

This lets us compare wildly different spreads on one common ruler: *"how stretched is this, relative to its own normal behaviour?"*

Mean reversion (the rubber-band idea)

Mean reversion is the observation that many relationships, after being stretched unusually far from their average, tend to snap *back* toward that average — like a rubber band. A spread that is +3 standard deviations "rich" (too high) often falls back toward normal; one that is −3 "cheap" (too low) often rises back.

The strategy in this project is a **mean-reversion** strategy. In one sentence:

> **When a spread becomes unusually stretched (a big z-score), bet that it will snap back toward normal.**

That's the whole idea. Everything in Phase 2 and Phase 3 is about doing this *carefully* — measuring "stretched" properly, only acting when the odds are good, and proving the idea actually works.

Part III — What the project is

The **Voltaire Energy Terminal** is a professional-style oil-trading dashboard. It was built in three phases, each adding a layer:

1. **Phase 1 — The live market dashboard.** Gather every relevant piece of live data (prices, inventories, weather, news, positioning) into one screen, so a trader can see the state of the market at a glance.
2. **Phase 2 — Market analysis & signal generation.** Go beyond just *showing* data: automatically read the market's current "mood" (its **regime**), work out what each spread *should* be worth (its **fair value**), and flag the spreads that look mispriced — ranked, scored opportunities.
3. **Phase 3 — The backtest.** Take the trading idea from Phase 2 and *test it on real market data*: simulate every trade it would have made, record the profit or loss of each, and judge whether the idea actually works.

The rest of this document explains each phase in financial terms.

Part IV — Phase 1: the live market dashboard

Phase 1 answers a simple question for the trader: **"What is happening in the oil market right now?"** It pulls together live, free, public data sources, each of which tells part of the story:

- **Prices & the forward curve** (from Yahoo Finance) — live futures prices for Brent, WTI and the refined products, and the shape of the forward curve (backwardation vs contango).
- **Inventories & fundamentals** (from the US EIA) — official weekly stockpile data: crude, petrol, diesel. The supply/demand truth.

- **Weather** (from Open-Meteo) — temperature forecasts in key demand regions; cold snaps drive heating-oil demand, heatwaves drive power/cooling demand.
- **Hurricanes/storms** (from the US NOAA) — storms in the Gulf of Mexico can shut down a large chunk of US oil production and refining, moving prices sharply.
- **Positioning** (from the US CFTC) — the weekly "Commitments of Traders" report, showing whether big speculative funds are heavily betting up or down. Extreme positioning often precedes reversals.
- **News & sentiment** (from a live newswire + an AI language model called FinBERT) — headlines are automatically scored as positive, negative or neutral for the oil market, giving a real-time read on the mood.

Individually, each is a data feed. Together, they are a **situational-awareness cockpit**: at a glance, is the market tight or loose, calm or stormy, bullishly or bearishly positioned?

This phase doesn't make trading decisions. It sets the stage for the phases that do.

Part V — Phase 2: reading the market's mood and finding mispriced trades

Phase 2 is where the project starts *thinking*. It has three jobs: classify the **regime**, compute each spread's **fair value**, and produce **ranked signals**.

1. Market regimes — the market's "mood"

A core insight of professional trading is that **the same trade behaves differently in different market environments.** Fading a stretched spread might be brilliant in a calm, well-supplied market and disastrous in a panicky, scarce one. So before anything else, the system classifies the *current environment* — its **regime**.

It does this by reading several independent dimensions of the market and labelling each:

- **Inventory** — are stockpiles Tight, Balanced, or Oversupplied? (from EIA data)
- **Term structure** — is the forward curve in Backwardation or Contango? (the scarcity signal)
- **Volatility** — is the market High, Normal, or Low vol?
- **Trend** — is oil trending Up, Down, or Range-bound?
- **Season** — are we in the Driving season (summer), Heating season (winter), or a shoulder period?

Combining these gives a compact label for the market's mood. In the data used for this project the live regime was, for example:

> **"Balanced · High Vol"** — inventories roughly at seasonal norms, but the market swinging violently (52% annualized volatility), with the forward curve backwardated (a tight, scarce signal) and prices trending up.

Why bother? Because the system then measures every spread's behaviour *within each regime separately*. "What is normal for the WTI calendar spread **when the market is Balanced and High-Vol**" is a much sharper question than "what is normal in general."

2. Fair value — what should this spread be worth?

For every spread, Phase 2 estimates a **fair value**: the price the spread *ought* to trade at, given current conditions. It does this two complementary ways:

1. **The regime average.** Within the current regime, what has this spread averaged historically? If WTI M1–M2 normally sits around 4.6 in a "Balanced · High Vol" regime, that's a baseline fair value.
2. **A statistical model (regression).** A more sophisticated estimate that relates the spread to underlying *drivers* — things like the US dollar's strength (DXY), the stock market's fear gauge (VIX), recent inventory changes, and price momentum. The model learns, from years of history, how the spread typically responds to these drivers, and outputs the value the spread *should* have given today's driver readings.

The **difference** between where a spread actually is and its fair value is the **dislocation** — the potential trade. A spread far *above* fair value is "**rich**" (a candidate to sell); far *below* is "**cheap**" (a candidate to buy).

The system expresses this dislocation as a **z-score** (Part II.7), so all spreads are measured on the same "how stretched is this?" ruler.

3. Ranked signals — the shortlist of opportunities

Phase 2 then produces a ranked list of **opportunities**, each carrying:

- **the instrument** (which spread) and **direction** (buy the cheap one / sell the rich one);
- **the dislocation** (the z-score — how stretched);
- **a written rationale** in plain English (e.g. "*Brent–WTI averages 6.26 in this regime; it is now 7.01, +0.2σ rich… backtest: such signals reverted 83% of the time*");

- **a confidence score**;
- and — critically — **a historical hit-rate from a validation backtest.**

4. The validation backtest — does this even work historically?

A trading idea is worthless unless it has actually worked. So Phase 2 includes a **historical validation**: it looks back over *years* of daily data and asks, for each spread, *"In the past, when this spread was stretched to 1.5 standard deviations, how often did it actually revert?"*

The answer — the **hit-rate** — is the foundation of trust in the strategy. For the spreads in this project, the historical hit-rates were strong:

Spread	Reverted (hit-rate)	vs random baseline	Edge
Brent–WTI	**83%**	49%	+34%
WTI butterfly	**82%**	49%	+32%
WTI M1–M2	**75%**	50%	+25%
Brent M1–M2	**72%**	48%	+24%

A "hit-rate" of 83% means: historically, when this spread was this stretched, it snapped back toward normal 83 times out of 100. The "baseline" (~49%) is what you'd expect from a coin-flip. The **edge** — the gap between them — is the evidence that the strategy is capturing something real, not luck.

This validation is *daily* data over *years*. Phase 3 is where we take this validated idea and run it on fresh, fast, intraday data.

Part VI — Phase 3: the backtest (testing the idea on real data)

Phase 2 proved the idea works on *years of daily* data. Phase 3 asks a tougher, more practical question:

> **"If I had actually run this strategy on real, minute-by-minute market data, trade by trade, exactly what would have happened?"**

This is a **backtest**: a careful simulation of the strategy on historical data, recording every single trade and its profit or loss. It is the closest thing to a dress rehearsal before risking real money.

This phase doesn't run *one* backtest — it runs **three**, on three different bodies of data, so the same idea is stress-tested from every angle:

1. **The live intraday backtest** — the strategy run on the **mentor's live company feed**, a fast 15-minute stream covering just the last few trading days. This is the "is it live right now?" check.
2. **The historical intraday backtest** — the same intraday method run over **5.4 years of 15-minute WTI/Brent bars**. This is the "does the fast signal hold up over thousands of trades and many market regimes?" check.
3. **The historical daily backtest** — the **Phase-2 fundamental model** run over **5.4 years of daily data**. This is the "does the slower, fundamentals-based version work out-of-sample?" check.

The first two share one method (the intraday engine); the third uses a different one (the daily fundamental model). We explain each in turn.

1. The data being tested

Two of the three backtests run on a database of **15-minute price bars** for crude futures. A "15-minute bar" simply summarizes all the trading in a 15-minute window (its open, high, low and closing price).

- The **live intraday** backtest uses the **mentor's live company feed**, which streams new bars as the market trades. As a *live* snapshot it is, by design, only a few days long.
- The **historical intraday** backtest uses **5.4 years of 15-minute WTI and Brent bars** (roughly 2021 → 2026) — the same method, but enough data to draw real statistical conclusions.

The third backtest, **historical daily**, runs on **5.4 years of daily** data, because the fundamental fair-value model it uses only updates daily (more on this below).

Because this data contains only crude (WTI + Brent), every backtest trades the crude spreads — calendars, butterflies, and Brent–WTI — and not the refined-product cracks.

2. The strategy being tested (restated precisely)

The Phase-2 idea — **relative-value mean-reversion** — but applied two different ways depending on the timeframe. The plain-English deep dive is in the companion document `Backtesting/STRATEGY.md`.

There is **one key rule** that ties the whole design together, so we state it up front:

> **The Phase-2 fundamental model is daily-only. Anything intraday must use a rolling, price-derived fair value instead.**

Why the intraday backtests cannot reuse Phase 2's fair-value model. Phase 2 built a clever statistical fair value for each spread from *fundamentals* (inventories, the dollar, volatility, the season). The natural instinct is to reuse it on the fast feed — but we looked carefully, and **it cannot be used intraday**, for three concrete reasons: - **Its ingredients aren't in this data.** The intraday feed contains *only oil prices* — none of the fundamental inputs the model needs — so it literally cannot be computed. - **Its number is out of date.** Phase 2's fair value for WTI Jul–Aug is ~**2.18**; that gap actually trades at ~**0.74** now. Forcing the old number on would flag a permanent false "cheap." - **It barely moves intraday.** Fundamentals update daily/weekly, so over a few hours the model is a flat line — it can't explain the minute-by-minute moves we're trading.

So the intraday method estimates fair value **from each spread's own recent history**, and keeps Phase 2 as *context* (its proven hit-rates feed the confidence score), not as the price reference. The daily method, run over years, *can* use the full fundamental model. We describe both.

(A) The intraday method — used by backtests 1 & 2 (live and historical intraday)

Step 1 — Estimate fair value. For each spread, take the **rolling average of its last 24 bars (~6 hours)**. That rolling mean is "normal." As new bars arrive it rolls forward, keeping up with a slowly drifting spread.

Step 2 — Measure the z-score. How far the spread is from that average right now, counted in *standard deviations* (its typical wiggle). A z-score of +2 means "two normal wiggles too high."

Step 3 — Enter on a clear dislocation *that is worth trading*. When the spread is stretched to **2 standard deviations or more**, consider fading it (sell if it's rich, buy if it's cheap) — but only **if it passes the cost gate.** - **The cost gate (the most important upgrade).** Before taking any trade, the engine asks: *is the expected dollar move back to fair big enough to be worth it?* It only enters if that expected capture clears roughly **2× a realistic round-turn trading cost.** Because a 2σ stretch is worth more dollars when the spread is volatile and almost nothing when it's calm, this gate is **volatility-adaptive**: it automatically **skips the cent-sized, low-volatility churn** — the trades that look fine gross but bleed money once you pay the bid–ask spread — and only fires on dislocations large enough to actually pay.

Step 4 — Exit by riding the reversion *through* fair, on the first of four triggers. - it reverts **through fair to a ~1.5σ overshoot on the other side** (the engine's `Z_EXIT = -1.5`) → **take profit.** This is a deliberate change from the old "exit at fair": spreads tend to *overshoot*, so closing exactly at fair left money on the table. Riding through to a 1.5σ overshoot captured more of each move and added net profit every year with no extra drawdown. - it stretches further against us to **3.5 standard**

deviations → stop loss; - it **hasn't resolved within ~12 hours → time stop** (cut the dead trades); - the session/weekend/roll break arrives → **flatten** (never hold blindly through a multi-day gap).

Pruning the universe — only the structures that survive costs. We evaluate all seven WTI and Brent crude structures, but we **only trade the three with a persistent, post-cost edge**: the **Brent–WTI arb**, the **WTI butterfly (M1-M2-M3)**, and the **WTI M2-M3 calendar**. The other four — notably the Brent *front* calendars — had no durable edge once costs were charged, in either the early or the late history, so they are dropped rather than carried as dead weight.

Sizing — fixed one unit per trade, with an optional scaling toggle. By default every trade is the same small size, so the result is the **raw per-unit signal**, not a leveraged book. There is an **optional 2-unit scaling toggle** that adds a second unit on the very deepest dislocations (≥2.75σ, the highest-conviction trades); it is **off by default** and reported separately so the headline figures stay clean. (An earlier version added 1%-of-capital sizing; it multiplied every position ~100× and turned the figures into scary six-figure numbers that obscured whether the *signal* was any good, so it was stripped back out. Sizing is a separate decision layered on *after* you trust the signal — see Part VII.)

(B) The daily method — used by backtest 3 (historical daily)

Run over **5.4 years of daily** data, the strategy *can* use the full Phase-2 machinery, so it does:

- **Fair value = the Phase-2 fundamental regression**, evaluated **walk-forward (out-of-sample)** — the model is only ever fitted on data *before* each trade, so it is never allowed to peek at the answer.
- **Enter** when the spread's residual (its gap from that fair value) is stretched to **≥1.5 standard deviations.**
- **Exit** at **0.5σ** (most of the snap-back), with a **3σ stop** and a **20-day time stop.**
- **Pruned to five structures** — the two losing **Brent front calendars** were dropped, leaving the five with a real out-of-sample edge.

This is the slower, fundamentals-grounded cousin of the intraday engine — and a direct out-of-sample test of the Phase-2 fair-value model itself.

3. The trade log — the main output

The centrepiece of Phase 3 is the **trade log**: a complete record of *every single trade* the strategy took. For each trade it records, in plain detail:

- **which spread** and **direction** (buy/sell);
- **entry time and price**, and the **z-score** at entry (how stretched it was);
- **exit time and price**, and **why** it exited (took profit at the overshoot / stopped out / **timed out** / session end);

- **how long it was held** (in bars and minutes);
- **the profit or loss in dollars** (the "PnL"), and in net mode the **cost** charged and the **net PnL** after it;
- **MAE / MFE** — the worst paper loss and best paper profit *during* the trade (how much heat it took before resolving);
- **the regime** it was traded in and a **confidence score** (grounded in Phase 2's validated hit-rate for that spread).

This log is produced both as a spreadsheet (`trades.csv`, openable in Excel) and as a human-readable document (`trades_log.md`) where each trade is written out in words.

What the dollar figures mean

Each oil contract is **1,000 barrels**, so a **$1.00 per barrel** move in a spread is worth **$1,000 per contract**. Every trade is a **fixed one unit**, so the dollar figures are the *raw per-unit signal* — small, honest, and directly comparable across trades. (The $250,000 starting capital is only a baseline for the running-total chart.) By default the backtest runs on a **gross basis with slippage set to zero** — measuring the *pure signal*, before real-world trading frictions; that was the brief's instruction, and it isolates the question *"does the prediction work?"*. Running it **net of realistic costs** (`--slip`) answers the separate, harder question *"can you actually execute it for a profit?"*.

4. The results

Each backtest is reported **net of a realistic per-leg cost** — the honest "can you actually execute it for a profit?" number — not just the flattering gross figure. The three together tell one consistent story.

Backtest 2 — Historical intraday (the headline result)

This is the big one: the intraday method over **5.4 years** of 15-minute bars, **net at 1¢/leg**, fixed one unit per trade.

Metric	Result	What it means
Trades	5,388	thousands of stretched-spread opportunities across 5+ years
Gross profit	**+$2,231,606**	total per-unit dollars made before costs
Net profit (1¢/leg)	**+$1,979,446**	after realistic costs — it **keeps ~89% of the gross**
Profit factor	**4.93**	for every $1 lost, $4.93 was won
t-statistic	**30.8**	overwhelmingly statistically significant (≈2 is the usual bar)
Per-year	**profitable every year 2021–2026**	not one good year carrying the rest
Max net drawdown	**≈ −$12,750**	the worst peak-to-trough dip along the way
With scaling ON	**net ≈ $2,622,400**	the optional 2-unit toggle on the deepest dislocations

How to read this:

- **The signal is strongly net-profitable, not break-even.** This is the headline change from the earlier version of this report. Because the **cost gate** skips the cent-sized churn and only fires on dislocations worth trading, the strategy **keeps ~89% of its gross profit after costs** — costs no longer eat the edge.
- **It is statistically real.** A profit factor near 5 and a **t-statistic of 30.8** over **5,388 trades** is not luck; the edge is durable, and it shows up in **every single year** from 2021 to 2026.
- **It is robust through drawdown.** The worst net drawdown across 5.4 years is about **−$12,750** — small relative to the ~$2.0M net it earned.

Backtest 3 — Historical daily (the fundamental model, out-of-sample)

The daily, Phase-2-fundamental method over the same **5.4 years**, **net at 2¢/leg**, walk-forward.

Metric	Result	What it means
Trades	198	far fewer, slower, higher-conviction trades
Gross profit	**+$117,183**	per-unit dollars before costs
Net profit (2¢/leg)	**+$94,143**	after costs — **keeps 80% of gross**
Profit factor	**2.34**	for every $1 lost, $2.34 was won
t-statistic	**3.4**	statistically significant (clears the ≈2 bar)
Per-year	**profitable all 6 years**	the fundamental model holds up out-of-sample

This matters because it is a **direct out-of-sample test of Phase 2's own fair-value model**: trained only on the past, judged on the future, and it is profitable every year. The two timeframes — fast price-derived and slow fundamentals-derived — agree.

Backtest 1 — Live intraday (the current few-day snapshot)

The same intraday method on the **mentor's live feed**, a snapshot only a few days long.

Metric	Result	What it means
Trades	~23	a handful of opportunities in a few live sessions
Gross profit	**~+$1,980**	per-unit dollars before costs
Net profit (1¢/leg)	**~+$820**	net-positive, but a small sample
Profit factor	**~2.8**	consistent with the historical edge

This is net-positive and consistent with the historical backtest, but with **only ~23 trades** it is a *spot-check that the live engine works*, not a statistical result — the confidence comes from the 5.4-year backtests above. The small-sample caveat (Part VII) applies to **this** snapshot only.

Part VII — How to read the results honestly

A good analyst is their own harshest critic. Here is the honest interpretation — exactly what you should tell a mentor or risk manager.

1. A high profit factor *and* statistical significance — why that's a good sign

The historical intraday strategy has a **profit factor of 4.93** — it won nearly $5 for every $1 lost — across **5,388 trades**. Crucially, that is paired with a **t-statistic of 30.8**, which means the result is overwhelmingly unlikely to be chance (the usual bar for "real" is about 2). Those two together are the healthy combination: the wins clearly outweigh the losses **and** there are enough of them, over a long enough history, to trust it. The reason the wins beat the losses is the **overshoot exit** — we ride each reversion **through fair to a ~1.5σ overshoot** rather than bailing at fair, so winners capture the full move, while losers are cut at the 3.5σ stop. A single number can be cherry-picked; a profit factor near 5 with a t-stat of 30.8, **profitable in every year from 2021 to 2026**, cannot.

The dollar totals are **per-unit on purpose**: one contract per trade. That is the *raw signal*, not a leveraged book — exactly what a clean test should show, and it is still strongly profitable.

2. The cost test — the verdict has flipped

This is the single biggest correction to the earlier version of this report. The old conclusion was: "*gross the signal works, but net of realistic fees it thins to roughly break-even, so costs decide it; the path forward is fewer, higher-conviction trades.*"

That has now been done — and it works. The upgraded method *is* exactly that fix: the **cost gate** only takes trades whose expected move clears ~2× a realistic cost (skipping the cent-sized churn), the universe is **pruned to the three structures with a persistent post-cost edge**, and the **overshoot exit** captures the full reversion. The result is that the strategy **keeps ~89% of its gross profit after costs** — net **+$1,979,446** against **+$2,231,606** gross over 5.4 years. The daily fundamental version keeps **80%** of gross net (net **+$94,143**). Costs no longer "decide it" against the strategy; the strategy was redesigned to beat them, and over 5.4 years it does — **profitable every single year.**

So the honest headline has reversed: this is **not** a break-even, costs-eat-the-edge story. It is a **strongly net-profitable, cost-aware, multi-year-validated** strategy.

3. Why we size at one fixed unit (and the optional scaling toggle)

The headline backtests size every trade at **one unit**, on purpose, so the result is the *raw signal*. There is an **optional 2-unit scaling toggle** that adds a second contract only on the deepest (≥2.75σ) dislocations; it is **off by default** and reported separately (with it on, historical-intraday net ≈ **$2,622,400**). An even earlier iteration tried a textbook **1%-of-capital sizing rule**; it multiplied every position by ~100×, turning the results into **six-figure swings** that said nothing new about the *signal* — they just scaled it up *and* scaled the risk up with it.

The lesson, kept here as the honest takeaway: **sizing is a separate decision you layer on *after* you trust the signal — and it needs portfolio-level risk limits, not just per-trade ones.** So we test the edge cleanly at one unit and expose scaling as an explicit, off-by-default option.

4. The pruned universe (and why it isn't cherry-picking)

The strategy does **not** trade every spread it can. The intraday engine evaluates all seven WTI and Brent crude structures but **trades only the three** with a persistent post-cost edge (Brent–WTI arb, the WTI butterfly, WTI M2-M3); the daily engine trades **five**, having dropped the two losing Brent front calendars. The fair concern is: *isn't picking the winners after the fact just curve-fitting?* The answer is the **walk-forward structure selection** in the robustness battery (next section): the structures are re-chosen each year using only **past** data, and the same three keep surviving — so the pruning is a rule the strategy could have followed in real time, not hindsight.

5. The fair-value caveat

The two timeframes deliberately use **different** fair values, and that is the **key rule** of the design: the **daily** backtest uses Phase 2's **fundamental regression** (walk-forward), while the **intraday** backtests use a **rolling 24-bar average of each spread's own recent prices** — because the fundamental model **can't be used intraday** (its inputs aren't in the feed, its stored level is stale, and it's flat over a few hours; see Part VI). The rolling average is the right tool for the fast timeframe, but it has an honest weakness: a plain mean can be **tugged by a single sharp spike**, slightly distorting "normal" on the very bar we react to. A more robust centre (a median) is a sensible refinement; we kept the simple average for clarity.

6. The robustness battery — the evidence the edge is real, not curve-fit

The strongest answer to "are you sure this isn't just a lucky backtest?" is the **robustness battery** (`analytics/robustness.py` → `server/data/robustness.json`), which stress-tests the historical edge five ways:

- **Per-year P&L** — the strategy is profitable in *every* year, not carried by one outlier.
- **t-statistic** — 30.8 (intraday) / 3.4 (daily): both clear the significance bar comfortably.
- **Monte-Carlo drawdown** — randomly reshuffling the trade sequence thousands of times to see how bad the drawdown *could* plausibly have been, so the ≈ −$12,750 figure isn't a lucky ordering.

- **Walk-forward structure selection** — re-picks the traded structures each year on past-only data, proving the 3-structure pruning isn't hindsight.
- **Parameter-sensitivity grid** — varies the thresholds (lookback, entry/exit z, cost) to confirm the edge isn't balanced on one magic setting.

Passing all five is what lets us call the edge **real and durable** rather than a curve-fit artefact.

7. The honest headline caveats, in one place

1. **The *live* sample is small.** The few-day live snapshot has only ~23 trades — a spot-check that the engine works, not a statistical result. **This caveat does *not* apply to the 5.4-year historical backtests** (5,388 and 198 trades), which are the basis for every conclusion above.
2. **Costs are illustrative.** The 1¢/leg (intraday) and 2¢/leg (daily) figures are realistic placeholders, not a specific desk's true bid–ask; they should be replaced with the desk's actual spread. (Net of them, the edge clearly survives — that is the point of §2.)
3. **15-minute closes aren't real fills.** The intraday backtest enters/exits at bar-close prices, which is an idealization of what you'd actually get in the book.
4. **Crude-only.** The data is WTI + Brent, so the product cracks aren't backtested here.
5. **One-unit raw signal.** The headline figures are the raw per-unit signal, not a sized/leveraged portfolio. Position sizing and portfolio risk limits are a separate layer on top (§3).

Stating these plainly is not weakness — it is exactly what separates a credible analysis from a misleading one.

Part VIII — How the three phases fit together

The project is a single, coherent pipeline that mirrors how a real trading desk thinks:

1. **See the market** (Phase 1) — gather every live signal: prices, the curve, inventories, weather, storms, positioning, news.
2. **Form a view** (Phase 2) — read the regime, compute fair values, and rank the mispriced spreads, each scored by a confidence grounded in years of validation.
3. **Prove the view** (Phase 3) — backtest the strategy trade-by-trade on real data, producing a complete, honest trade log with full profit-and-loss accounting.

Each phase feeds the next. The regime and the validated hit-rates from Phase 2 flow directly into the confidence score on every trade in Phase 3. The whole thing is designed to move from *information* → *insight* → *evidence*.

Glossary — every term in one line

- **Barrel** — the standard unit of oil; one futures contract = 1,000 barrels.
- **Backwardation** — forward curve slopes down (near oil dearer than far); signals scarcity/tightness.
- **Benchmark** — a reference grade of oil; the two main ones are Brent and WTI.
- **Brent** — the North Sea / global benchmark oil (code `CO`, ICE exchange).
- **Brent–WTI spread** — the price gap between the two benchmark oils.
- **Butterfly / fly** — a 3-leg spread (+1 M1, −2 M2, +1 M3) betting on the curvature of the forward curve.
- **Calendar spread** — the price gap between two contract months of the same oil (e.g. M1–M2).
- **Contango** — forward curve slopes up (near oil cheaper than far); signals oversupply/glut.
- **Contract month / tenor** — the delivery month of a futures contract, coded by a letter + year.
- **Cost gate** — a volatility-adaptive entry filter: only trade if the expected move to fair clears ~2× realistic cost; skips cent-sized low-vol churn.
- **Crack spread** — the gap between crude and the refined fuels made from it; a refiner's margin proxy.
- **Drawdown** — the drop from a peak in account value to a later trough; a measure of pain/risk.
- **EIA** — US Energy Information Administration; publishes official weekly oil inventory data.
- **Expectancy** — the average profit per trade across all trades.
- **Fair value** — what a spread *should* be worth, given conditions; the reference for "rich/cheap."
- **Fade** — to bet *against* a recent move (sell something that just rose, buy something that just fell).
- **Forward curve / term structure** — the line of futures prices across all delivery months.
- **Front month (M1) / prompt** — the nearest-to-expiry, most-traded contract.
- **Futures contract** — a standardized agreement to buy/sell oil at a set price for future delivery.
- **Gross basis** — results before trading costs (slippage, commission).
- **Hedger** — a market participant trading to reduce risk (airlines, refiners, producers).
- **Hit-rate** — the % of times a stretched spread historically reverted; the core measure of edge.

- **Net basis** — results *after* trading costs (slippage); the honest "tradeable profit" figure, vs the gross "pure signal."
- **Time stop** — closing a trade that hasn't resolved within a set time (~12h here), to avoid dead positions.
- **Inventories / stocks** — oil sitting in storage; the clearest read on supply vs demand.
- **Leg** — one of the individual contracts inside a spread trade.
- **Liquidity** — how easily a contract can be traded without moving its price.
- **MAE / MFE** — Maximum Adverse / Favourable Excursion: the worst/best paper PnL during a trade.
- **Mean reversion** — the tendency of a stretched value to snap back toward its average.
- **Monte-Carlo drawdown** — reshuffling the trade order thousands of times to estimate how bad the drawdown could plausibly have been.
- **Overshoot exit** — taking profit not at fair but after the spread reverts *through* fair to a ~1.5σ overshoot, capturing the full move.
- **PnL** — Profit and Loss; the money made or lost.
- **Profit factor** — total dollars won ÷ total dollars lost; above 1 is profitable.
- **Pruning** — trading only the structures with a persistent post-cost edge (3 intraday, 5 daily) and dropping the rest.
- **Regime** — the market's current "mood," labelled across inventory, structure, volatility, trend, season.
- **Regression** — a statistical model relating a spread to its underlying drivers, to estimate fair value.
- **Rich / cheap** — a spread trading above (rich) or below (cheap) its fair value.
- **Slippage** — the small cost of trading (the gap between expected and actual fill price); zero in gross mode, configurable for the net test.
- **Speculator** — a participant trading to profit from price moves, not to hedge.
- **Spread** — the price difference between two related contracts, traded as one position.
- **Standard deviation** — the "typical wiggle" of a value; the unit behind the z-score.
- **Stop loss** — a pre-set exit that caps the loss on a losing trade.
- **t-statistic** — a measure of how unlikely a result is to be chance; above ~2 is "statistically significant."
- **Volatility** — how violently a price swings; usually an annualized %.
- **Walk-forward / out-of-sample** — fitting a model only on past data and testing it on later, unseen data, so it can't peek at the answer.
- **WTI** — the US benchmark oil (code `CL`, CME/NYMEX exchange).

- **z-score** — how far a value is from its average, measured in standard deviations; the "how stretched?" ruler.

End of report. This document covers the project end-to-end: the financial foundations, the live dashboard (Phase 1), the regime analysis and signal generation (Phase 2), and the trade-by-trade backtest with its honest interpretation (Phase 3).